Satellite Image Land-Use Classification and Temporal Change Detection Using Transfer Learning

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I. INTRODUCTION

Satellite imagery provides critical information for applications such as environmental monitoring, urban planning, precision agriculture, and disaster management. The increasing availability of high-resolution Earth observation data has created a growing demand for automated image analysis methods capable of extracting meaningful geographic information at scale. Deep learning, particularly convolutional neural networks combined with transfer learning, has significantly improved the accuracy and efficiency of land-use classification tasks while reducing the need for handcrafted feature engineering.

This project presents an end-to-end computer vision pipeline for satellite image land-use classification and temporal change detection. A ResNet18-based model is fine-tuned on the EuroSAT dataset to classify satellite imagery into ten land-use categories. The learned feature representations are subsequently reused to detect temporal land-cover changes through cosine similarity between image embeddings. To improve evaluation reliability, the dataset is partitioned using geographically aware spatial clustering, reducing spatial leakage between training and testing data. The complete system is deployed as a Streamlit application for interactive inference and visualization.

II. PROBLEM STATEMENT

Traditional random train-test splits often overestimate model performance because neighboring satellite tiles may share similar geographic characteristics, resulting in spatial leakage. Furthermore, classification models trained on a single dataset may not generalize well to imagery collected from different sensors or geographic regions. This project addresses these challenges by combining spatially aware evaluation, external validation on the UC Merced dataset, and embedding-based temporal change detection within a unified deep learning framework.

III. OBJECTIVE

This project aimed to develop a robust satellite image analysis pipeline by implementing an accurate land-use classification model based on transfer learning and employing geographically aware train-validation-test splitting to reduce spatial leakage. Additionally, the project sought to evaluate model generalization using the UC Merced holdout dataset, perform temporal land-cover change detection through deep feature embeddings and cosine similarity, and integrate these capabilities into an interactive Streamlit dashboard for real-time inference and visualization.

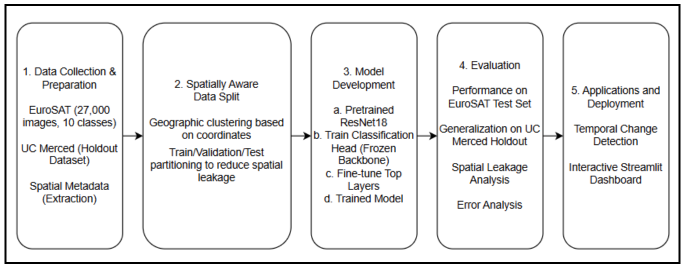

Fig 1: Project Pipeline

A. Class Distribution:

The dataset is reasonably balanced, with class sizes ranging from 2,000 to 3,000 images, reducing the need for aggressive class-balancing strategies.

Class	**Images**
AnnualCrop	3000
Forest	3000
HerbaceousVegetation	3000
Highway	2500
Industrial	2500
Pasture	2000
PermanentCrop	2500
Residential	3000
River	2500
SeaLake	3000

Table 1: Class Distribution Table

B. Preprocessing

Images were resized from 64 × 64 pixels to 224 × 224 pixels to match the input requirements of ResNet18. Pixel values were normalized using dataset-specific mean and standard deviation values computed over the complete EuroSAT dataset:

```
Mean = [0.3444, 0.3803, 0.4078]
Std  = [0.2027, 0.1369, 0.1155]
```

Data loaders were constructed using PyTorch with separate training, validation, and testing datasets to ensure reproducible experimentation.

C. Spatial Metadata

Unlike conventional image classification datasets, EuroSAT provides corresponding multispectral Sentinel-2 imagery containing geographic coordinate information. The multispectral GeoTIFF files were used to extract the geographic center (latitude and longitude) of each image tile. These coordinates enabled spatial clustering and geographically aware dataset partitioning while the RGB images were used for model training.

D. Spatially Aware Data Split

Rather than employing a random train-test split, the dataset was partitioned using geographic clustering based on image coordinates. Images belonging to the same spatial cluster were assigned entirely to either the training, validation, or testing set, thereby reducing spatial leakage between splits. This evaluation strategy provides a more realistic estimate of model performance on unseen geographic regions.

Split	Images
Training	21,568
Validation	2,774
Test	2,658

Table 2: Splitting of data into train-val-test subsets

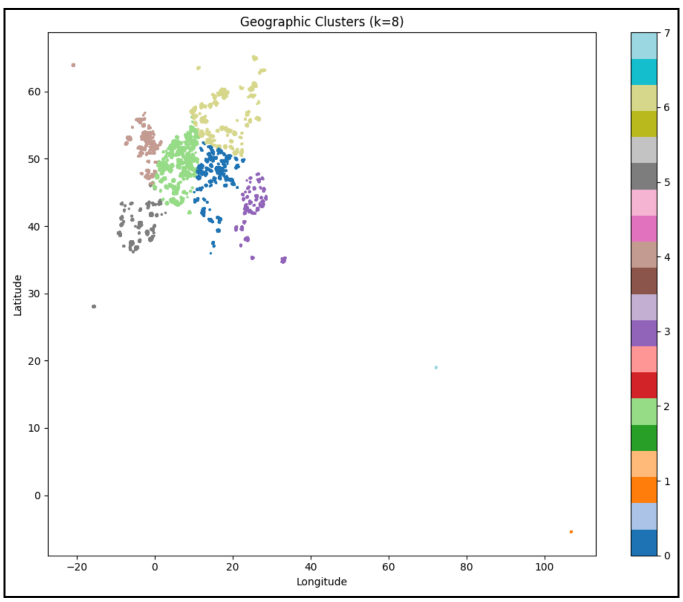

Fig 2: Geographic cluster visualization

IV. MODEL DEVELOPMENT

A. Baseline CNN

A custom convolutional neural network (CNN) was initially developed to establish a baseline for satellite land-use classification. The model consisted of three convolutional layers followed by fully connected classification layers and was trained entirely from scratch on the EuroSAT dataset. This baseline provided a reference for evaluating the effectiveness of transfer learning using deeper pretrained architectures.

B. Transfer Learning with ResNet18

To improve classification accuracy and training efficiency, a ResNet18 model pretrained on ImageNet was adopted as the primary backbone. Instead of training all network parameters from scratch, transfer learning enabled the model to leverage rich visual representations learned from large-scale natural image datasets and adapt them to satellite imagery.

During the first stage, all convolutional layers of the pretrained ResNet18 backbone were frozen, and only the final fully connected classification layer was optimized. This allowed the classifier head to learn the EuroSAT land-use categories while preserving the pretrained feature representations.

After the classifier head converged, selected layers of the ResNet18 backbone were unfrozen and jointly optimized using a lower learning rate. Fine-tuning enabled the model to adapt higher-level visual features to the characteristics of satellite imagery while retaining the benefits of pretrained weights.

C. Training Strategy

All models were trained using the same data split, optimizer, batch size, learning rate, and early stopping criterion. The best-performing checkpoint based on validation loss was used for evaluation.

Parameter	**Baseline CNN**	**Frozen ResNet18**	**Fine-tuned ResNet18**
Optimizer	Adam	Adam	Adam
Loss Function	CrossEntropyLoss	CrossEntropyLoss	CrossEntropyLoss
Batch Size	64	64	64
Learning Rate	0.001	0.001	0.001
Weight Decay	1e-4	-	-
Maximum Epochs	30	20	15
Early Stopping	Patience = 5	Patience = 5	Patience = 5
Model Checkpointing	Yes	Yes	Yes

Table 3: Hyperparameters

D. Training Techniques

Cross-entropy loss was used as the optimization objective for all experiments. The Adam optimizer was employed throughout training, while early stopping with a patience of five epochs prevented overfitting by monitoring validation loss. The model weights corresponding to the best validation performance were automatically saved through checkpointing for subsequent evaluation.

V. EXPERIMENT RESULTS

Three models were evaluated on the spatially partitioned EuroSAT test set: a baseline CNN, a frozen ResNet18 model, and a fine-tuned ResNet18 model. Transfer learning significantly improved classification performance compared to training a CNN from scratch, demonstrating the effectiveness of pretrained feature representations for satellite imagery.

Model	**Accuracy**	**Macro F1**
Baseline CNN	57.67%	56.19%
Frozen ResNet18	74.79%	74.51%
Fine-Tuned ResNet18	74.79%	74.51%

Table 4: Comparison Table of Accuracy and Macro F1 of Baseline CNN, Frozen and Fine-Tuned ResNet18 models on EuroSAT test data.

Here, we observe that transfer learning improved accuracy by approximately 17% over the baseline CNN. Fine-tuning achieved performance comparable to the frozen model in the reported evaluation, while requiring additional optimization of higher-level feature representations.

VI. GENERALIZATION

To assess model robustness, the trained models were evaluated on the UC Merced Land Use dataset without additional fine-tuning. Since UC Merced contains imagery collected from different geographic regions and acquisition conditions, it provides a realistic assessment of domain generalization.

Model	**Accuracy**	**Macro F1**
Baseline CNN	17.86%	9.27%
Frozen ResNet18	30.93%	20.96%
Fine-tuned ResNet18	44.29%	33.50%

Table 5: Comparison Table of Accuracy and Macro F1 of Baseline CNN, Frozen and Fine-Tuned ResNet18 models on UC Merced Land Use Dataset.

Although performance decreased compared with EuroSAT, the fine-tuned ResNet18 substantially outperformed the baseline CNN, indicating that the learned representations retained meaningful semantic information despite the domain shift between datasets.

Split Strategy	**Accuracy**	**Macro F1**
Spatial Block Split	74.79%	74.51%
Random Stratified Split	88.72%	88.36%

Table 6: Comparison Table of Accuracy and Macro F1 of Spatial Block Split and Random Stratified Split Strategy.

VII. ERROR ANALYSIS

Error analysis was performed to better understand the limitations of the fine-tuned classifier. Misclassified samples, prediction confidences, and class-wise performance were examined to identify recurring failure patterns. Most classification errors occurred between visually similar vegetation-related classes, particularly PermanentCrop, HerbaceousVegetation, and AnnualCrop. These categories exhibit comparable spectral and textural characteristics, making them difficult to distinguish using RGB imagery alone. In contrast, visually distinctive classes such as Forest, Residential, and SeaLake achieved consistently high precision and recall.

	filepath	true_label	predicted_label	true_class	predicted_class	confidence	correct
0	c:\Users\ASUS\dev\projects\satellite-project\d...	6	2	PermanentCrop	HerbaceousVegetation	0.999993	False
1	c:\Users\ASUS\dev\projects\satellite-project\d...	3	6	Highway	PermanentCrop	0.999941	False
2	c:\Users\ASUS\dev\projects\satellite-project\d...	4	7	Industrial	Residential	0.999890	False
3	c:\Users\ASUS\dev\projects\satellite-project\d...	6	2	PermanentCrop	HerbaceousVegetation	0.999794	False
4	c:\Users\ASUS\dev\projects\satellite-project\d...	8	0	River	AnnualCrop	0.999766	False

Fig 3: Examples of incorrect predictions illustrating confusion between visually similar land-use categories.

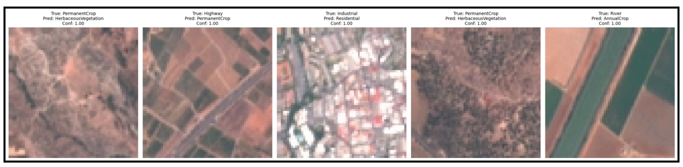

Fig 4: Five representative high-confidence misclassifications produced by the fine-tuned ResNet18 model. These examples highlight common failure modes arising from similar vegetation textures, mixed land-use scenes, and limited spatial resolution.

VIII. TEMPORAL CHANGE DETECTION

A. Methodology

To extend the capabilities of the land-use classification model, the fine-tuned ResNet18 network was repurposed as a feature extractor for temporal change detection. The final fully connected classification layer was removed, allowing the network to generate 512-dimensional feature embeddings that capture the semantic content of each satellite image. These embeddings provide a compact representation suitable for measuring similarity between images acquired at different time periods.

Temporal image pairs were generated by grouping images according to their geographic clusters and simulating "before" (T1) and "after" (T2) observations. For each image pair, cosine similarity was computed between the extracted embeddings. Image pairs with high similarity were considered unchanged, whereas pairs with lower similarity were classified as changed. An optimal similarity threshold was selected using Receiver Operating Characteristic (ROC) analysis.

B. Performance Evaluation

The embedding-based change detector demonstrated strong discrimination between changed and unchanged image pairs. ROC analysis produced an Area Under the Curve (AUC) of **0.9461**, indicating excellent separability of the two classes. The optimal cosine similarity threshold was determined using Youden's Index, resulting in a threshold of **0.7274**, corresponding to a True Positive Rate (TPR) of **0.90** and a False Positive Rate (FPR) of **0.15**.

Using this threshold, the change detection model achieved an overall classification accuracy of **87.42%** on the generated evaluation set. The balanced precision and recall values for both classes indicate that the embedding representations learned during land-use classification effectively capture semantic differences associated with temporal land-cover changes.

Metric	**Value**
ROC AUC	0.9461
Cosine Similarity Threshold	0.7274
Detection Accuracy	0.8742
True Positive Rate	0.90
False Positive Rate	0.15

Table 7: Performance metrics of the embedding-based temporal change detection model.

C. Qualitative Analysis

To further validate the proposed approach, representative image pairs were visualized together with their corresponding difference heatmaps. The generated heatmaps clearly highlight regions exhibiting significant visual changes, while unchanged image pairs maintain high embedding similarity and minimal pixel-wise differences. These qualitative results demonstrate that the learned feature embeddings are capable of identifying meaningful land-cover changes without requiring a dedicated segmentation model.

Overall, the results show that deep feature embeddings extracted from a fine-tuned classification network can be effectively reused for temporal change detection, providing an efficient and computationally lightweight solution for monitoring changes in satellite imagery.

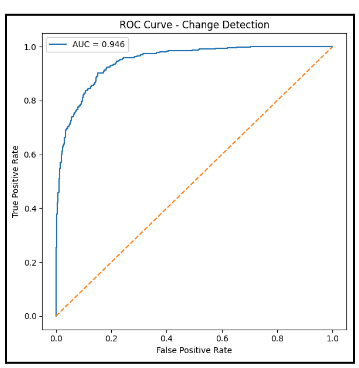

Fig 5: ROC Curve - Change Detection

IX. DASHBOARD

A Streamlit-based dashboard was developed to provide an interactive interface for satellite image analysis. The application integrates the trained classification model and the embedding-based change detector into a single workflow, allowing users to upload satellite images and obtain predictions in real time.

X. LIMITATIONS

Despite achieving strong performance, the proposed system has several limitations. First, the land-use classification model relies solely on RGB satellite imagery, whereas multispectral Sentinel-2 data could provide additional spectral information for improved discrimination between visually similar classes. Second, temporal change detection was evaluated using simulated before-and-after image pairs generated from spatially partitioned data rather than true multi-temporal satellite observations, which may not fully capture real-world land-cover changes.

Third, although external validation on the UC Merced dataset demonstrated reasonable generalization, the model was evaluated on a limited number of datasets and may require further adaptation for imagery acquired from different sensors, resolutions, or geographic regions. Finally, the use of ResNet18 offers a good balance between accuracy and computational efficiency but may be outperformed by more recent architectures such as Vision Transformers or foundation models designed specifically for remote sensing applications.

XI. CONCLUSION

This project developed a complete satellite image analysis pipeline integrating land-use classification, spatially aware evaluation, temporal change detection, and interactive deployment. A baseline CNN established an initial performance benchmark, while transfer learning with ResNet18 substantially improved classification accuracy. Geographic clustering enabled realistic train-validation-test partitioning, reducing spatial leakage and providing more reliable evaluation. The trained feature extractor was further utilized for embedding-based temporal change detection using cosine similarity, demonstrating the versatility of deep feature representations beyond classification. Finally, the entire workflow was deployed as a Streamlit application for interactive inference and visualization.